

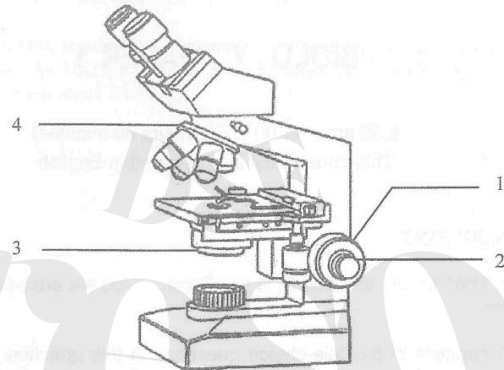
1. 2014Q5

A student wants to use an ordinary light microscope to observe the binary fission of a photosynthesizing protist under high magnification. A temporary mount of the protist is placed on the stage of the microscope. Which of the following structures would be observable in the above study?

- A. Ribosome
- B. Chloroplast
- C. Mitochondrion
- D. Endoplasmic reticulum

2. 2015Q1

Which of the following parts of the microscope should be adjusted to obtain a clear and sharp image when you switch from low-magnification to high-magnification observation?



A.

- B. 1 and 4 only
- C. 2 and 3 only
- D. 1, 3 and 4 only
- E. 2, 3 and 4 only

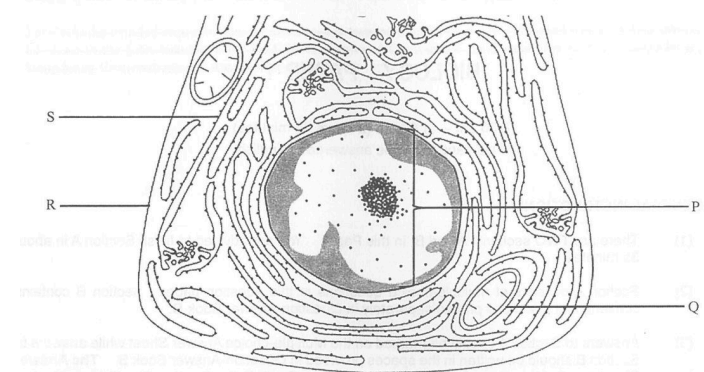
3. 2020Q1

Which of the following sub-cellular structures can be found in a prokaryotic cell?

- A. cell wall, ribosome, cell membrane
- B. ribosome, cell membrane, chloroplast
- C. cell wall, cell membrane, mitochondria
- D. cell membrane, chloroplast, mitochondria

4. 2023Q1

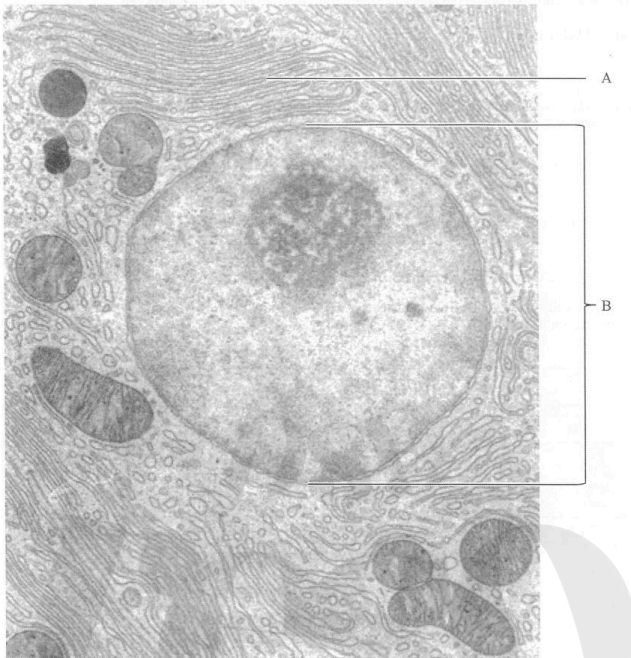
Question 1 refer to the schematic diagram below, which shows part of a pancreas cell:



Which of the following cellular components can also be found in a prokaryotic cell?

- A. P
- B. Q
- C. R
- D. S

LQ
1. 2017Q4a

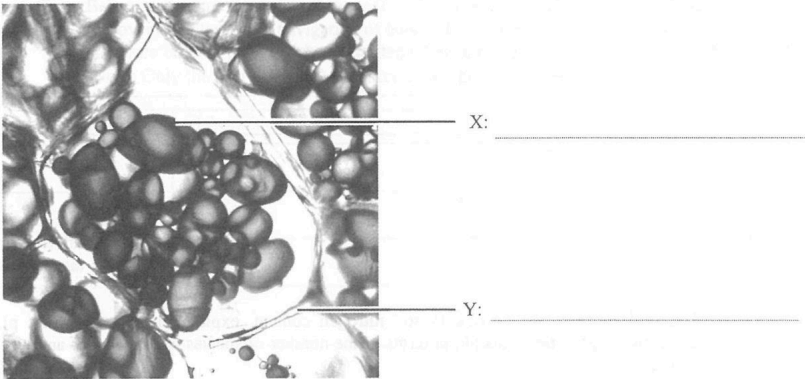


The electron micrograph shows some structures of a human cell
a) Label A and B.

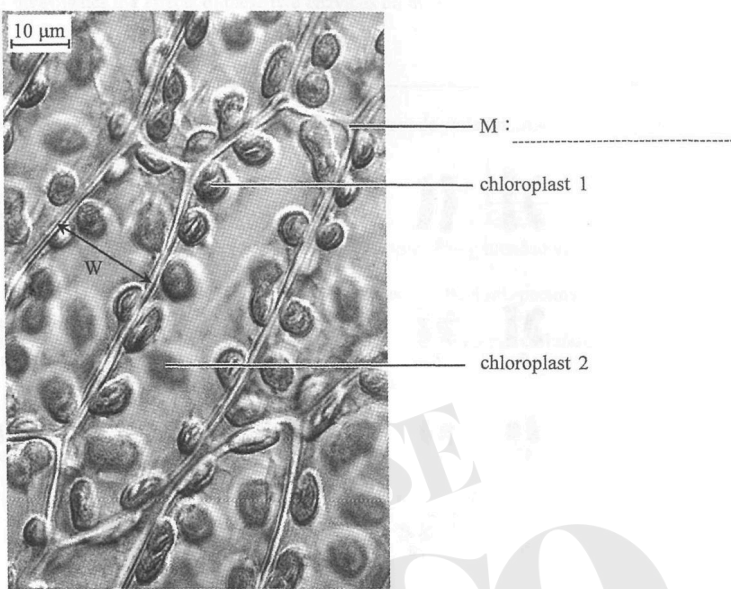
(2 marks)

2. 2019Q6d
Johnny prepared a slide of freshly sectioned potato cylinder and stained it with iodine solution
The photomicrograph below shows the section.
d) Label structures X and Y.

(2 marks)



3. 2022Q3
The photomicrograph below shows some unstained plant cells:



- a) Label structure M. (1 mark)
- b) What is the actual length of W shown in the photomicrograph? (1 mark)
- c) Chloroplast 1 appears sharp in this photomicrograph while chloroplast 2 appears blurred. To obtain a sharper image of chloroplast 2, how should you operate the microscope? (1 mark)
- d) Some structures of the chloroplast cannot be distinguished in this photomicrograph. State **one** of these structures. (1 mark)
- e) Give **one** equipment that can be used to observe the structure stated in (d). (1 mark)

Ans

2014Q5: B (41%)

2015Q1:D (49%)

2020Q1:A (63%)

2023Q1:C (64%)

2017Q4a

A: endoplasmic reticulum (1)

B: nucleus (1)

2019Q6d

X: starch granule / starch grain (1)

Y: cell wall (1)

2022Q3

a) M: cell wall (1)

b) 19 (± 1) μm (1)

no mark if no unit is given

c) rotate the fine adjustment knob until the image is in focus (1)

d) granum / thylakoid membrane / inner membrane / outer membrane (1)

e) electron microscope (1)

DSE
I'eso